	Why Are We Doing This Project? (Meaning & Purpose)	The hiring project is a "reality-mimicking simulation" designed to put you directly in the driver's seat of a production-level challenge. Rather than evaluating you on abstract textbook questions or memorized algorithms, this project simulates your "Worst/Hardest Day" on the job. It tests how you translate vague specifications into secure, operational, and mathematically clean cloud architectures while protecting our data integrity. We operate without micromanagement. We must verify that you can analyze a problem, leverage documentation, and build robust technical solutions independently. In our development pipelines, we enforce absolute structural discipline. We do not rely on "hope" or "human memory" to prevent bugs; we build automated "mistake-proofing" (Poka-Yoke) systems. This project assesses whether you can design automated systems that enforce our operational "Golden Rules"
	What Will Be the Outcome of the Project?	Your project will produce a Polished Deployment & Automation Blueprint consisting of modular Terraform scripts, automated pipeline templates, and structured data-validation models. For us, the outcome is undeniable, objective evidence of your capacity to provision GCP projects securely, establish automated build gates, and map clean data flows
	Must know	Hiring at HabotConnect is a demanding process. Typically 1 out of 180+ candidates are accepted. You must show that you know your work. If you wish to be a HaboTech, you must know its: 1. Values 2. Leadership Principles
	Project Context	HabotConnect is building a digital platform that connects parents with Learning Support Assistants (LSAs) for children with learning difficulties. To scale this platform securely, our backend Django REST Framework (DRF) and frontend React applications deploy via automated pipelines to Google Cloud Platform (GCP). We require an entry-level engineer who can translate deconstructed business logic into secure, cost-efficient cloud resources with zero deviation from our security protocols
	You Will Be Assessed On	1. **Infrastructure-as-Code (IaC) Design:** Your ability to provision GCP App Engine and databases securely and cleanly using Terraform. 2. **Poka-Yoke (Mistake-Proofing) Pipeline Logic:** Implementing automated linter, formatting, and security build gates that block non-compliant code from deploying. 3. **Data Pipeline & Schema Validation:** Ensuring transactional

		schemas align with Pub/Sub and BigQuery streaming sinks without data loss. 4. **Identity & Access Control:** Configuring strict role-based access controls (RBAC) and following Least Privilege principles. 5. **Technical Rigor and Clarity:** Meticulous documentation, zero reliance on placeholders, and absolute adherence to structural rules.
	You are Expected To Do	**You are presented with a critical staging scenario:** A junior developer has pushed an update that bypassed security guidelines, leaving unencrypted API credentials in raw application code, and has triggered a database schema mismatch that broke down-stream analytics. **You must execute the following 3 tasks to restore system integrity:** 1. **Task 1: Terraform Secure Staging Provisioning (IaC):** Write a Terraform configuration (`main.tf`) to securely provision a Google Cloud Storage (GCS) raw landing bucket (D0 Raw Landing) and a BigQuery dataset (D1 Staged/Enforced). Apply strict IAM conditions and Row-Level Security (RLS) policies. **Task 2: Poka-Yoke Automated CI/CD Build Gate:** Design a YAML configuration (e.g., GitHub Actions) that executes automated linter checks and security scans. Your pipeline **must act as a strict "Fail-Closed" gate**—it must immediately fail, halt, and quarantine the build if formatting errors or raw, hardcoded API secret keys are detected. **Task 3: Schema Mapping and DCYN Validation:** Deconstruct an incoming JSON payload (representing a student onboarding form) into a binary Yes/No logic library (DCYN library). Write a clean Django REST Framework model serializer class with exact field validation limits to entirely eliminate human judgment.
	The Presentation / Submission Is To Be	**Instructions to Complete:** **You are required to:** 1. Deliver your completed Terraform, YAML, and Python code blocks in a structured folder layout. 2. Present the engineering logic behind your automated linter gates and data pipelines. 3. **Your presentation/submission should include the following:** 4. Present your architectural overview and logic flow using a **Google Slides or PowerPoint presentation (Maximum of 15 slides)**.

		5. Demonstrate how your automated build gate successfully triggers a "Fail-Closed" status on invalid or insecure commits. 6. All spreadsheet worksheets used for data/schema mapping must have **"Wrap Text" enabled** for full visibility, and you must **Use Full Forms Only** (no slang, abbreviations, or placeholders). 7. Ensure your answer document and code files are clearly labeled with your full name and contact information at the top 8. **Submission Link (GoogleForm):** https://forms.gle/u5LFdyH2KWds5CvE6 **Submission Deadline:** 3 August 2026
	Suggestions & Guidelines	• **Timeframe:** The project should take you 4 to 6 hours maximum to complete. • **Resources:** You are expected to use outside resources to help you understand what is required (e.g., basic platform logic principles). • **Independence:** Show us that you can do what is required by us without supervision. No company resource person will be available to assist you. • **Self-Screening:** Use this opportunity to find out whether you truly want to work within HabotConnect's highly accountable, detail-obsessed, "Quiet Management" remote culture.
	When	You will present your completed project during the Project Interview phase.
	Where	Video call presentation via Google Meet. You will receive a calendar invite with the date and time from our Human Resources team members.
	Who	You will be presenting to panel members which may include the CEO, Team Lead, and Human Resources team members.
	How	• It is assumed that you will let us know if you are unable to do the required presentation and that you no longer wish to continue with the interview process at HabotConnect. Your slide presentation should include links to your English Code documents or sheets if required. • Slide presentation with links to other documents or sheets if required. Max of 15 slides
	Additional HabotConnect Resource links	**The Leadership Principles:** Leadership Principles -HAW 3 2025-271124.pdf **HabotConnect Values:** Habot- Values-HAW3-291124.pdf